

Data Requirement Specification

1. Objective

This document defines the financial, macroeconomic, market-risk, and planned news-sentiment data required for model development and empirical analysis.

2. Required Dataset

Category	Dataset	Description	Source	Frequency	Period
Financial	JPM Stock Prices	Daily historical prices of JPMorgan Chase & Co.	Yahoo Finance/ Alpha Vantage	Daily	2018–2024
Macroeconomic	U.S. Treasury Yield	Proxy for the risk-free interest rate used in option pricing	FRED	Daily	2018–2024
Market risk indicator	CBOE Volatility Index (VIX)	Expected market volatility and risk conditions; not news sentiment	Yahoo Finance	Daily	2018–2024
News sentiment	Timestamped JPM news sentiment score	Planned daily 0–1 feature with publication-time controls	Reuters News API / NewsAPI, subject to licensed historical access	Daily	2018–2024 where available

3. Scope Definition

This project uses the Black–Scholes framework. The data specification therefore covers the underlying asset, the risk-free-rate proxy, market volatility conditions, and a planned news-sentiment input.

Financial: Daily JPM stock prices represent the underlying asset of the chooser option.

Macroeconomic: The U.S. Treasury yield serves as the risk-free-rate proxy.

Market risk: VIX reflects expected equity-market volatility and risk conditions. It is not a news-sentiment score.

Collection status: the supplied project environment did not provide a reproducible licensed historical news archive. Weeks 2–8 therefore disclose the missing sentiment series and do not relabel VIX as sentiment.